3D CLOCK TOWER

A 3D Architectural Visualization in OpenGL

PROJECT OVERVIEW

Our project aimed to design and build a realistic 3D clock tower, complete with intricate architectural details and a dynamic environment.

CORE CONCEPT

Create a detailed clock tower model as a prominent landmark.

KEY FEATURES

An animated clock face and dynamic day-night cycle with lighting and shadows.

ENVIRONMENTAL INTEGRATION

The tower is set within a complete 3D scene, featuring roads, buildings, and natural elements

IMPLEMENTATION & Modeling Techniques

PRIMITIVE SHAPES

Constructing the tower and scene elements using fundamental shapes like cubes, spheres, and cones.

PROCEDURAL TEXTURING

Simulating brick patterns and other surface details for enhanced realism without external textures.

ADVANCED LIGHTING

Implementing directional light for the sun, and point lights for street lamps, with day/night cycles.

SHADOW MAPPING

Generating dynamic shadows cast by the clock tower to add depth and realism to the scene.

ALPHA BLENDING

Creating transparent effects for clouds, fog, and light halos for a more atmospheric rendering.

THE CLOCK TOWER: ARCHITECTURAL DETAIL

1.CIRCULAR STEPPED BASE: A strong foundation providing a sense of impressiveness.

2.MAIN TOWER BODY: Textured with a varying brick pattern, giving it an authentic feel.

3.ARCHED WINDOWS: Replicated on all four sides, adding structural elegance.

4.FUNCTIONAL CLOCK FACE: Displays real-time with hour, minute, and second hands.

5.DECORATIVE STONE BANDS: Horizontal accents breaking the verticality of the tower.

6.BELL TOWER/CUPOLA SECTION: An open, arched structure near the top.

7.PYRAMID ROOF & FINIAL: A classic pyramidal roof crowned with a decorative spire.

DYNAMIC ENVIRONMENT: DAY & NIGHT CYCLE

The scene transitions between day and night, affecting lighting, fog, and the activation of street lamps

DAYTIME SCENE

Bright sun with realistic rays, light blue sky, and subtle atmospheric haze.

NIGHTTIME SCENE

Darker sky gradient, denser fog, and illuminated street lamps with glow effects.

SCENE COMPOSITION: A BUSTLING JUNCTION

The clock tower is set within a dynamic urban intersection, enhancing the realism and scale of the model.

- **Intersecting Roads:** North, Southeast, and Southwest roads creating a three-way junction.
- **Road Markings & Crossings:** Detailed center lines and a zebra crossing for authenticity.
- **Street Lamps:** Strategically placed at junction corners, providing illumination at night.
- **Traffic Lights:** A functional traffic light adds to the urban environment.
- **Animated Cars:** Multiple cars positioned on roads, some in motion, some waiting.
- **Surrounding Buildings:** A variety of buildings line the roads, contributing to the cityscape.
- **Sidewalks & Greenery:** Sidewalks with segments and scattered trees add natural elements.

CHALLENGES FACED

- Designing a realistic 3D clock tower structure
- Implementing natural sunlight and lighting effects
- Creating a realistic sky and environment
- Managing camera movement and viewing angles
- Proper placement of buildings and roads

NEW KNOWLEDGE & Future Improvements

New Knowledge Gained

- Experience with geometric modeling using primitive shapes
- In-depth understanding of lighting models, shadow projection, and blending techniques.
- Principles of creating dynamic and interactive 3D environments.

Possible Improvements

- Adding complex textures using image mapping for greater detail.
- Improving vehicle movement and interaction logic.
- Introducing real-time weather effects like rain or snow.
- Expanding the city environment with more diverse buildings and landscapes.